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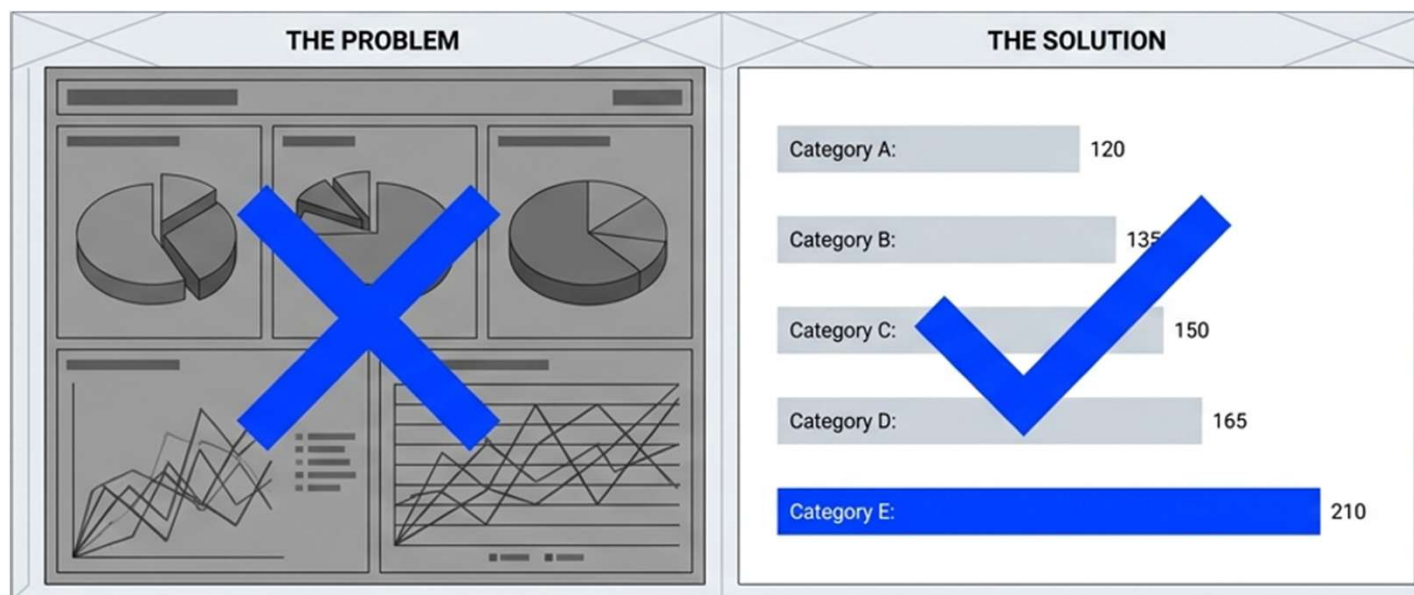
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Rules for data visualization



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Bad Visualization is a Crisis of Trust, not Aesthetics

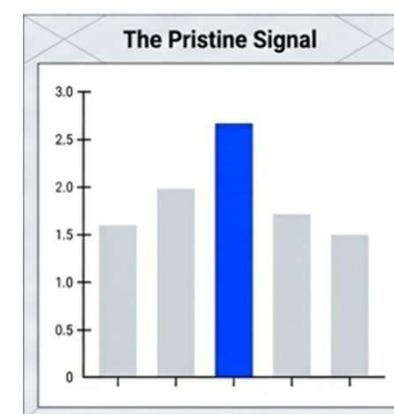
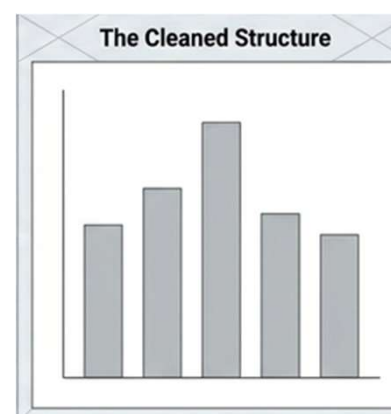
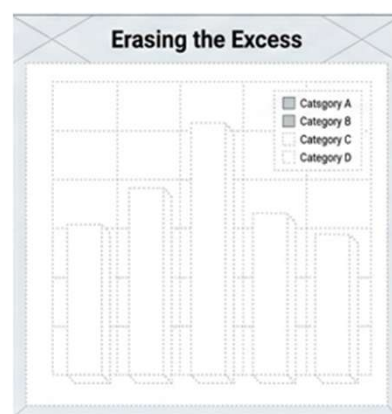
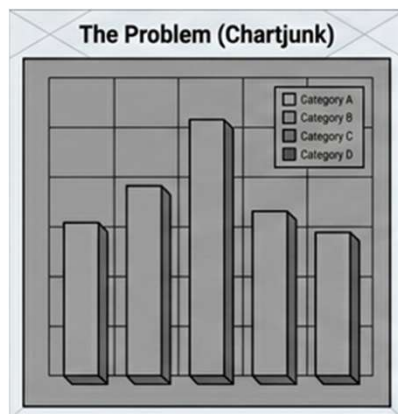


When visual clarity breaks down, users stop trusting the product. Poor design doesn't just waste space—it hides patterns, shifts attention from what matters, and leads stakeholders to entirely wrong conclusions.

Erasing the noise: the Data-Ink Ratio

- Edward Tufte's golden rule: Maximize the proportion of ink that represents actual data. Every pixel must server the data

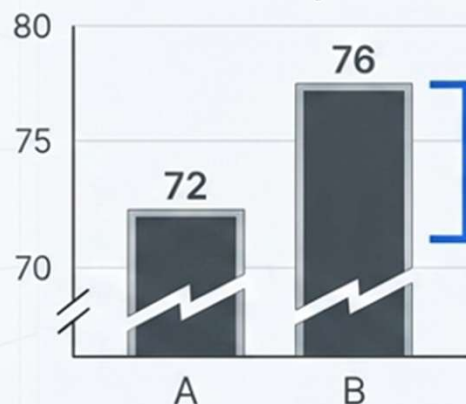
$$\text{Data-Ink Ratio} = \frac{\text{Data-Ink}}{\text{Total Ink Used}}$$



The Lie Factor: Quantifying Visual Deception

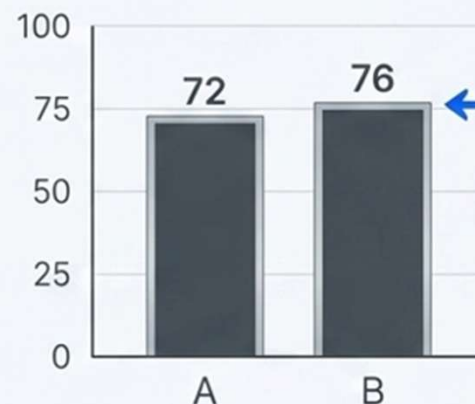
$$\text{Lie Factor} = \frac{(\text{Size of effect in graphic})}{(\text{Size of effect in data})} = 36x$$

The Deception



Graphic Change:
Bar height
increases by 200%

The Truth

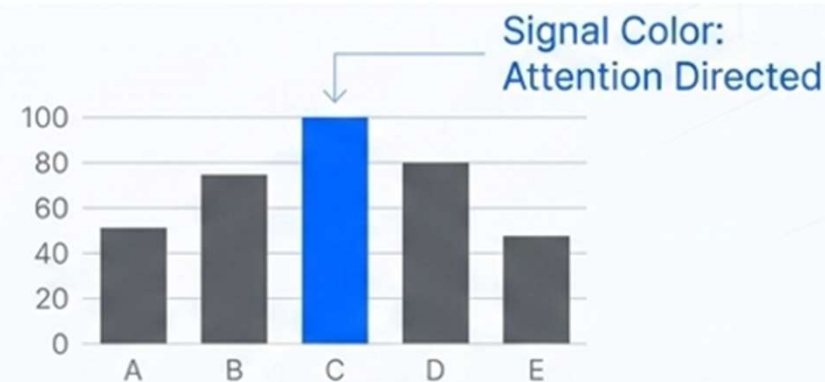


Data Change:
72 to 76 is a
5.6% increase

Result: 200% / 5.6% = 36x Lie Factor

Color is for Meaning, not Decoration

Color for Meaning



The visual felony diagnostic grid



The chart selections decision matrix

Goal	Best Choice	Use with Caution	Do Not Use
Show Change Over Time	Line Chart  Small Multiples 	Area Chart (can cause occlusion) 	Pie Chart 
Compare Categories	Horizontal Bar  Dot Plot 	Grouped Column (bad for small screens) 	Radar / Spider Chart 
Show Parts of a Whole	Stacked Bar 	Donut Chart (max 3 slices) 	3D Pie Chart 
Show Correlations	Scatter Plot 	2D Histogram 	Dual-Axis Line Charts 



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Thank You!



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